CUSTOMER CHURN ANALYTICS REPORT

AfriTel Ghana — Telecom Churn Intelligence Study

End-to-End Analysis: Python → SQL → Power BI

PREPARED BY

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DATASET

Telecom Customer Churn Dataset (Kaggle) — adapted as AfriTel Ghana business simulation

DATE

[April 2026]

2,666	388	14.55%	$64.84	214
Total Customers	Churned Customers	Churn Rate	Avg Churned Revenue	High Risk Customers

1. EXECUTIVE SUMMARY

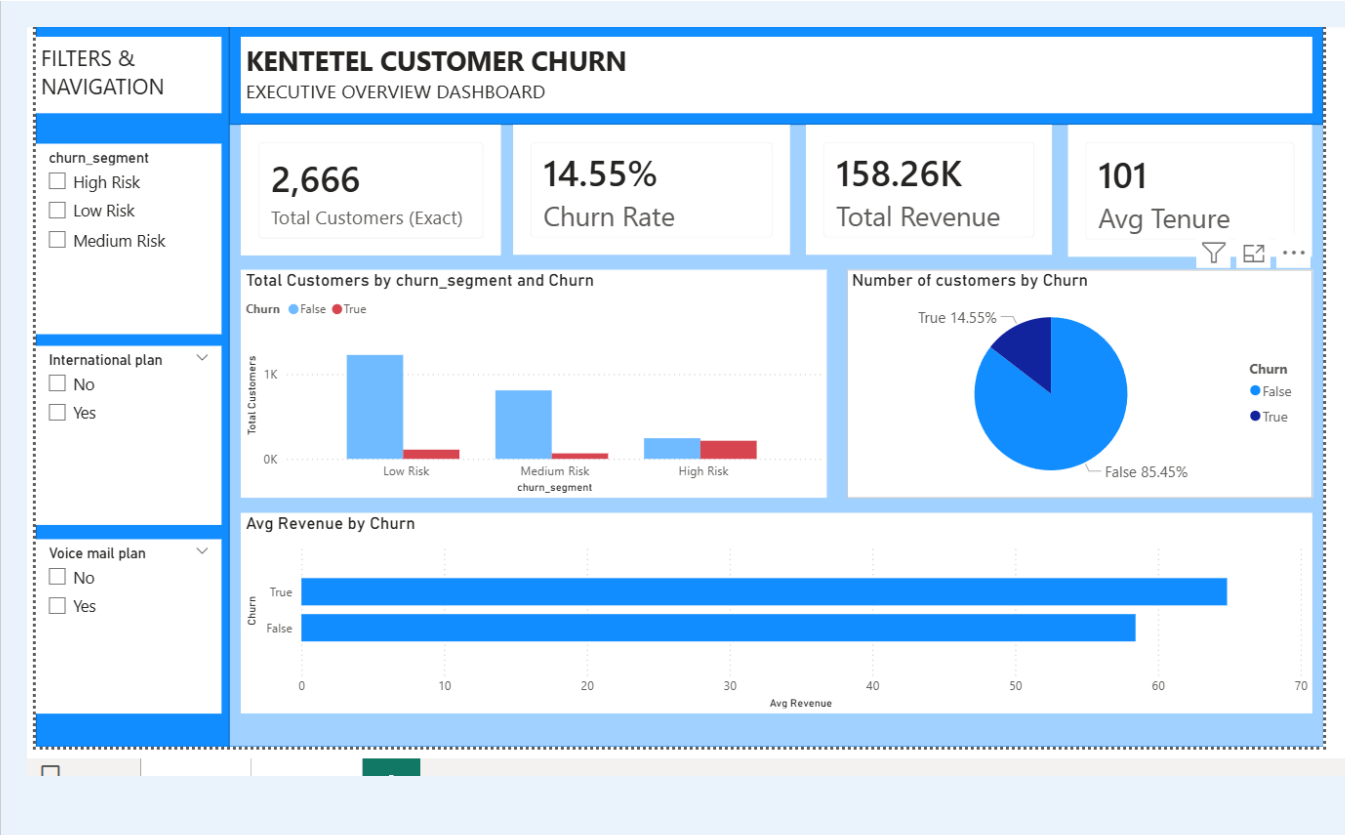
This analysis evaluates customer churn behaviour at **AfriTel Ghana**, a fictional mid-size Ghanaian telecom operator modelled on real transactional data across **2,666 customers** and **388 confirmed churn cases**. AfriTel Ghana faces a **14.55% churn rate** driven by three critical business threats:

01	**Service Failure Driving Churn -** customers making 4+ service calls churn at rates between 48% and 100%.
02	**International Plan Dissatisfaction -** 43.7% churn rate among international plan users vs 11.27% for non-users.
03	**High-Value Customer Loss -** churned customers generate $64.84 average revenue vs $58.43 for retained customers.

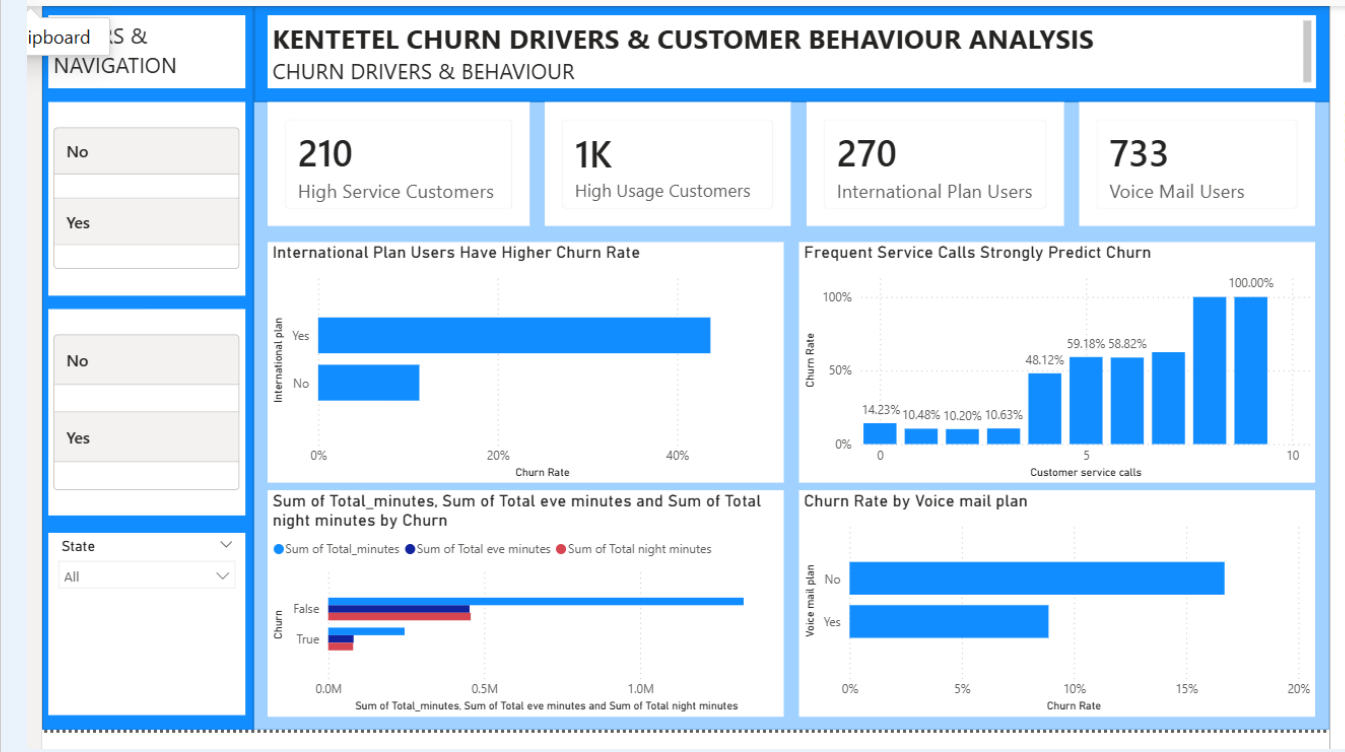
AfriTel Ghana's churn problem is not a usage problem - it is a **service experience problem**. Churned customers use the network MORE than retained customers (1,045 mins vs 983 mins), making every churn case a preventable loss of a high-engagement, high-revenue subscriber.

Dashboards

Dashboard 1 - Executive Overview



Churn Drivers and Behaviour



2. BUSINESS OBJECTIVE

AfriTel Ghana engaged this analysis to answer five critical business questions:

- Why are customers leaving despite high usage - and what are the primary churn triggers?
- Which customer segments carry the highest churn risk and what differentiates them?
- What is the true revenue cost of current churn levels to the business?
- Is the international plan a commercial asset or a retention liability?
- What specific, data-backed interventions can AfriTel Ghana deploy in the next 90 days to reduce churn?

3. DATA PREPARATION AND METHODOLOGY

Data was extracted, cleaned, and validated using Python and SQL, followed by segmentation modelling and visualisation in Power BI.

3.1 Data Cleaning

- Handled missing values and standardised categorical columns across all fields
- Validated churn labels and confirmed binary classification integrity
- Derived churn risk segments using service call frequency and plan type combinations

3.2 Feature Engineering

- **Churn Segment** - High Risk / Medium Risk / Low Risk classification based on service calls and plan type
- **Revenue Differential** - Comparison of average revenue between churned and retained customers
- **Usage Analysis** - Total minutes comparison across churn status to test usage-as-retention hypothesis

3.3 KPIs Defined

KPI	Purpose
Churn Rate	Overall subscriber loss percentage
Avg Revenue - Churned	Revenue value of lost customers
Avg Revenue - Retained	Benchmark for retention value comparison
Service Call Churn Trigger	Call count at which churn rate accelerates past 48%
International Plan Churn	Plan-specific churn rate differential vs base rate
High Risk Segment Size	Volume of customers requiring urgent intervention

3.4 SQL Queries Used

The following queries validated key findings extracted during Python analysis.

Customer_Churn_Analysis.ipynb PYTHON

Python — Step 1: Load Data and Initial Exploration

```python
import pandas as pd

# Load AfriTel Ghana customer dataset
df = pd.read_csv('churn-bigml-80.csv')

# Create clean working copy — never modify raw data
df_clean = df.copy()

# Verify dataset dimensions: 2,666 rows × 20 columns
print(df_clean.shape)

# Confirm no missing values across all features
df_clean.info()

# Churn rate: True = 14.55%, False = 85.45%
df_clean["Churn"].value_counts(normalize=True) * 100
```

Customer_Churn_Analysis.ipynb PYTHON

Python — Step 2: Feature Engineering (Total Usage & Risk Score)

```python
# Engineer Total_minutes feature — sum of all usage dimensions
df_clean["Total_minutes"] = (
    df_clean["Total day minutes"] +
    df_clean["Total eve minutes"] +
    df_clean["Total night minutes"] +
    df_clean["Total intl minutes"]
)

# Risk scoring function — weighted by churn predictors
def churn_risk(row):
    score = 0
    if row["International plan"] == "Yes":
        score += 2                      # high-risk product
    if row["Customer service calls"] >= 4:
        score += 3                      # strongest churn predictor
    elif row["Customer service calls"] >= 2:
        score += 1
    if row["Total_minutes"] > df_clean["Total_minutes"].mean():
        score += 1                      # high usage indicator
    return score

df_clean["Risk_Score"] = df_clean.apply(churn_risk, axis=1)
```

Customer_Churn_Analysis.ipynb PYTHON

Python — Step 3: Risk Level Segmentation & Export

```python
# Map risk scores to segment labels
def risk_level(score):
    if score >= 4:   return "High Risk"
    elif score >= 2: return "Medium Risk"
    else:            return "Low Risk"

df_clean["Risk_Level"] = df_clean["Risk_Score"].apply(risk_level)

# Segment distribution
print(df_clean["Risk_Level"].value_counts())
# High Risk:   458
# Medium Risk: 873
# Low Risk:    1335

# Churn rate by segment confirms model accuracy
df_clean.groupby("Risk_Level")["Churn"].mean()
# High Risk:   0.467  (46.7% churn)
# Medium Risk: 0.074  (7.4%  churn)
# Low Risk:    0.082  (8.2%  churn)

# Export for Power BI
df_clean.to_csv("churn_powerbi_ready.csv", index=False)
```

churn.db | SQLite via Python SQL

SQL — Query 1: Churn Rate by Service Call Volume

```sql
-- Business question: At what call threshold does churn
-- rate accelerate — revealing the intervention trigger point?

pd.read_sql("""
SELECT
    "Customer service calls",
    COUNT(*)                                        AS total_customers,
    SUM(CASE WHEN Churn = 1 THEN 1 ELSE 0 END)      AS churned,
    AVG(CASE WHEN Churn = 1 THEN 1 ELSE 0 END) * 100 AS churn_rate_pct
FROM churn_data
GROUP BY "Customer service calls"
ORDER BY "Customer service calls"
""", conn)

-- Result: churn jumps from ~14% to 48%+ at call 4
-- reaching 100% at calls 8 and 9
```

churn.db | SQLite via Python

SQL

SQL — Query 2: Revenue Comparison — Churned vs Retained Customers

```sql
-- Business question: Are churned customers higher or lower value
-- than retained customers — and what is the total revenue at risk?

pd.read_sql("""
SELECT
    Churn,
    COUNT(*)                    AS total_customers,
    AVG("Total day charge"
        + "Total eve charge"
        + "Total night charge"
        + "Total intl charge")  AS avg_revenue,
    SUM("Total day charge"
        + "Total eve charge"
        + "Total night charge"
        + "Total intl charge")  AS total_revenue
FROM churn_data
GROUP BY Churn
""", conn)

-- Result: Churned = $64.84 avg  |  Retained = $58.43 avg
```

churn.db | SQLite via Python

SQL

SQL — Query 3: International Plan Churn Rate Analysis

```sql
-- Business question: Is the international plan a churn accelerator
-- and by how much does it elevate risk vs non-users?

pd.read_sql("""
SELECT
    "International plan",
    COUNT(*)                                        AS total_customers,
    SUM(CASE WHEN Churn = 1 THEN 1 ELSE 0 END)       AS churned,
    AVG(CASE WHEN Churn = 1 THEN 1 ELSE 0 END) * 100 AS churn_rate_pct
FROM churn_data
GROUP BY "International plan"
""", conn)

-- Result: Plan users = 43.7% churn
--         Non-users  = 11.27% churn  (3.9x difference)
```

churn.db | SQLite via Python SQL

SQL — Query 4: Customer Risk Segmentation with Churn Rate & Revenue

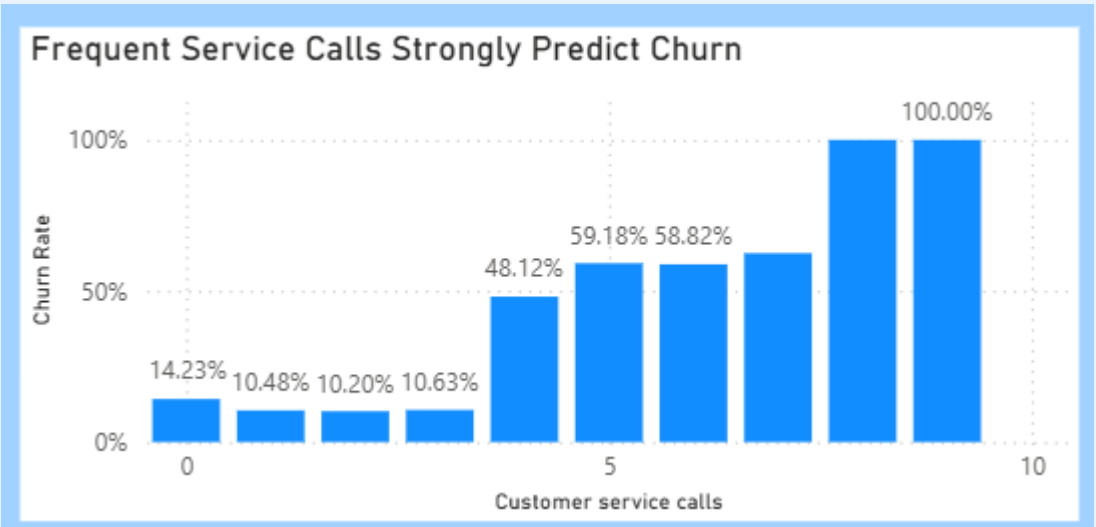
```sql
-- Business question: Which customer segments carry the highest
-- churn risk and what is their average revenue contribution?

pd.read_sql("""
SELECT
    CASE
        WHEN "Customer service calls" >= 4
             OR "International plan" = 'Yes'
             OR ("Total day minutes" + "Total eve minutes"
                + "Total night minutes" + "Total intl minutes") >
                (SELECT AVG("Total day minutes" + "Total eve minutes"
                          + "Total night minutes" + "Total intl minutes")
                 FROM churn_data)
        THEN 'High Risk'
        WHEN "Customer service calls" >= 2
        OR ("Total day minutes" + "Total eve minutes"
           + "Total night minutes" + "Total intl minutes") >
           (SELECT AVG("Total day minutes" + "Total eve minutes"
                     + "Total night minutes" + "Total intl minutes")
            FROM churn_data)
        THEN 'Medium Risk'
        ELSE 'Low Risk'
    END AS churn_segment,
    COUNT(*) AS total_customers,
    AVG(CASE WHEN Churn = 1 THEN 1 ELSE 0 END) * 100 AS churn_rate,
    AVG("Total day charge" + "Total eve charge"
       + "Total night charge" + "Total intl charge") AS avg_revenue
FROM churn_data
GROUP BY churn_segment
""", conn)
```

4. ANALYSIS AND KEY FINDINGS

4.1 Customer Service is the Primary Churn Driver

Bar Chart - Service Calls vs Churn Rate

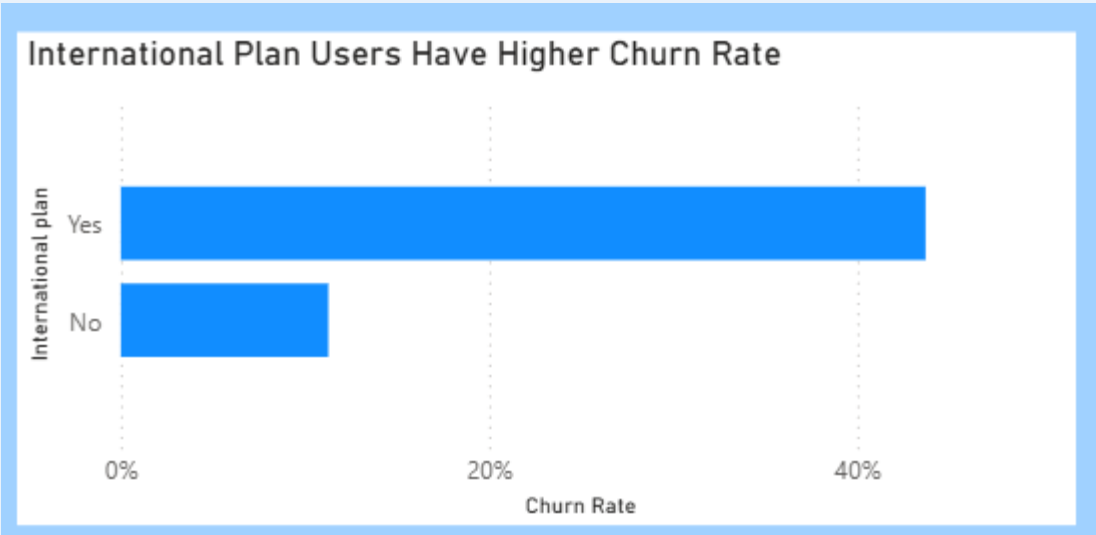


INTERPRETATION

Churn crosses the **48% threshold at exactly 4 service calls** and reaches **100% churn by the 9th call**. Every contact after the 3rd call signals a failure to resolve - not engagement. AfriTel Ghana currently has **6 customers with 9+ service calls**, all of whom have churned. The 3rd service call is the critical intervention point - the last moment at which retention is still achievable.

4.2 International Plan is a High-Risk Product

Bar Chart - International Plan vs Churn Rate



INTERPRETATION

International plan users churn at **43.7%** - nearly **four times the rate** of non-users at 11.27%. AfriTel Ghana has **270 international plan subscribers**. At the current churn rate, approximately **118 of these customers are at risk**. At $64.84 average revenue per churned customer, this represents **$7,651 in monthly revenue at risk** from the international plan segment alone. The plan is not delivering sufficient value to justify its retention risk.

4.3 High Usage Does NOT Prevent Churn

Churned Customers	Retained Customers
1,045 mins avg usage	**983 mins avg usage**

INTERPRETATION

This is the most counterintuitive and strategically important finding. Churned customers use the network MORE than retained ones - by 62 minutes on average. This definitively rules out inactivity as the cause of churn and points entirely to **service experience failure**. AfriTel Ghana is not losing disengaged customers. It is losing its most active, highest-value subscribers. Standard engagement-based retention models will not work here - the problem requires a service quality solution, not a usage incentive programme.

4.4 Revenue Loss - AfriTel is Losing High-Value Customers First

$64.84	$58.43	$6.41	$25,182
Avg Revenue - Churned	Avg Revenue - Retained	Revenue Gap per Customer	Monthly Revenue Lost

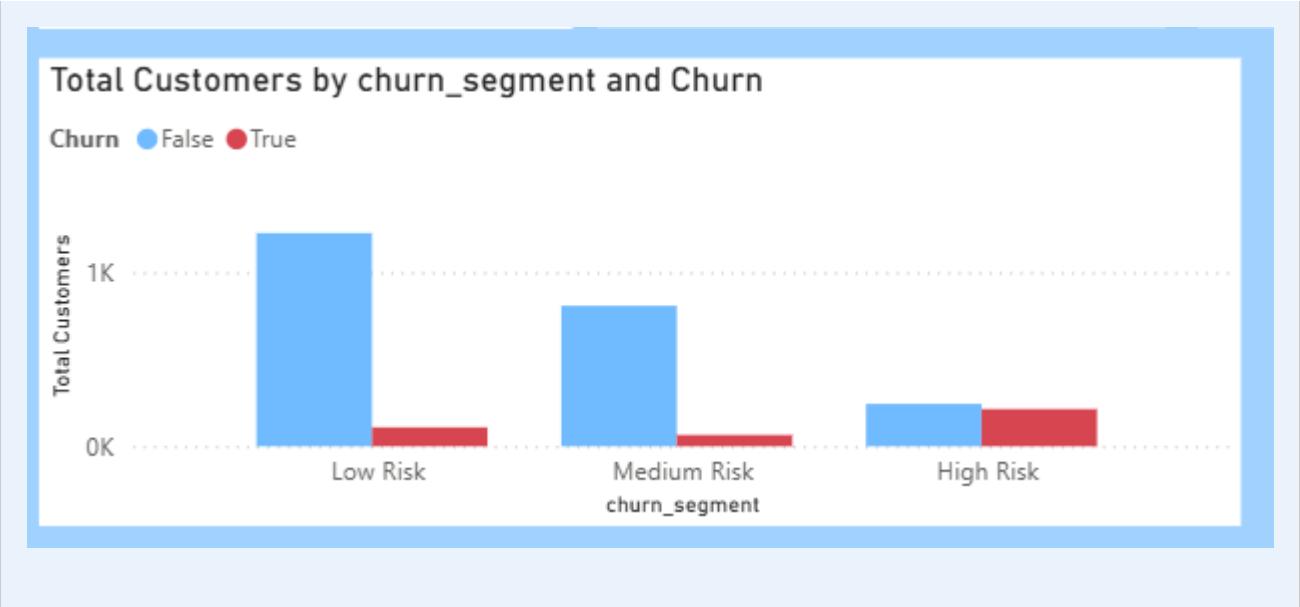
INTERPRETATION

At 388 churned customers averaging $64.84 per month, AfriTel Ghana is losing approximately **$25,182 in monthly recurring revenue** - equivalent to **$302,189 in annualised revenue loss**. The $6.41 revenue gap confirms that churn is **disproportionately affecting high-value subscribers**. Left unaddressed, the customer base gradually shifts toward lower-value retained customers while high-value subscribers exit - permanently eroding AfriTel Ghana's revenue ceiling.

4.5 Customer Risk Segmentation

HIGH RISK	MEDIUM RISK	LOW RISK
214 churned 46.7% churn rate	**7.4% churn rate** Moderate intervention	**8.2% churn rate** Monitor and maintain

Segmentation Chart - Customers by Risk Tier



INTERPRETATION

The **214 High Risk customers** represent AfriTel Ghana's most urgent retention priority. With a 46.7% churn rate this segment is almost coin-flip likely to leave. Retaining just 50% of the High Risk segment would protect approximately **$6,950 in monthly recurring revenue**. The Medium Risk segment (7.4%) represents a prevention opportunity - intervening before these customers escalate to High Risk is significantly cheaper than attempting to recover churned customers.

5. KEY BUSINESS RISKS

!	**Critical**	Service call churn trigger - 48% to 100% churn for customers with 4+ calls. No early intervention system currently in place.
!	**Critical**	$25,182 monthly revenue lost - equivalent to $302,189 annualised. Accelerating as high-value customers exit first.
!	**High**	International plan churn at 43.7% - nearly 4x the base rate. Plan is acting as a churn accelerator, not a retention tool.
!	**High**	High-value customers churning first - $64.84 avg vs $58.43 retained, compressing future revenue ceiling.
!	**Medium**	No proactive retention programme - all intervention is reactive. By the time churn is detected, customers have already decided to leave.

6. RECOMMENDATIONS

1 Implement the 3-Call Intervention Rule

The data shows churn crosses the **48% threshold at exactly 4 service calls**. AfriTel Ghana must implement a proactive callback protocol triggered automatically when any customer reaches their 3rd service call - assigning a senior agent and prioritising resolution within 24 hours.

At current volumes, retaining just 5 high-risk customers per month generates $324 in protected monthly revenue - exceeding the cost of one dedicated retention intervention. The business case pays for itself within 30 days of implementation.

2

Conduct International Plan Emergency Review

With 270 subscribers churning at 43.7%, the international plan is AfriTel Ghana's single highest-risk product. A structured review must address: Is the pricing competitive? Are customers experiencing quality issues? Is the value proposition clear at point of sale?

Immediate action: Survey all 270 international plan subscribers. Offer a **30-day satisfaction guarantee** - if not satisfied, downgrade with no penalty. This proactive gesture costs less than the $7,651 monthly revenue currently at risk from international plan churn.

3

Build an Early Warning Churn Scoring System

AfriTel Ghana currently has no mechanism to identify at-risk customers before they leave. The data provides three clear early warning signals: **service call frequency**, **international plan subscription**, and **high usage combined with complaint history**.

A basic scoring model flagging any customer with 3+ service calls OR international plan subscription as High Risk would have correctly identified the majority of the 388 churned customers before they left. Deploying this as a weekly automated report allows the retention team to act proactively.

4

Launch Targeted High-Risk Retention Campaigns

The **214 High Risk customers** require immediate personalised outreach. Service-dissatisfied customers receive a **priority support upgrade**; international plan users receive a **rate review or loyalty credit**.

Retaining 50% of the High Risk segment (107 customers) at $64.84 average revenue protects **$6,938 in monthly recurring revenue** - a 27.5% reduction in current monthly churn losses achievable within 90 days.

5

Establish Monthly Churn Intelligence Reporting

AfriTel Ghana currently has no structured churn monitoring cadence. A monthly Power BI dashboard tracking churn rate by segment, service call volumes,

international plan performance, and revenue at risk gives leadership the visibility needed to act before churn escalates.

Key monthly metrics: overall churn rate vs 14.55% baseline, High Risk segment size, international plan churn rate, average service calls per churned customer, and total monthly revenue at risk. Trend lines over 6 months reveal whether interventions are working.

7. CONCLUSION

AfriTel Ghana's 14.55% churn rate is not a volume problem - it is a **service quality and product experience problem**. The data is unambiguous: churned customers use the network more, pay more, and leave because their problems are not being resolved fast enough. The company is actively losing its best customers to preventable service failures.

The path forward is clear and quantifiable. **Implementing the 3-call intervention rule** addresses the primary churn trigger. **Reviewing the international plan** removes the highest-risk product from its current churn-accelerating role. **Deploying an early warning scoring system** shifts AfriTel Ghana from reactive to proactive retention. Together, these three interventions could protect an estimated **$6,938 in monthly recurring revenue** - a 27.5% reduction in current monthly churn losses achievable within 90 days.

All Python notebooks and SQL queries are available in the project GitHub repository.

GitHub: https://github.com/emmanuella-danso